Experiment 10

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Branch: CSE - AIML
Section/Group: 24AIT_KRG-G1
Semester: 4
Date of Performance: 24/04/26
Subject Name: DBMS
Subject Code: 24CSH-298

1. Aim

To design a trigger that automatically implements the functionality of a primary key, ensuring unique identification of records without manual intervention.

2. Software Requirements

- Database Management System:
  - Oracle Database Express Edition (Oracle XE)
  - PostgreSQL Database
- Database Administration Tool / Client Tool:
  - Oracle SQL Developer (for Oracle XE)
  - pgAdmin (for PostgreSQL)

3. Objectives

To create a database trigger that automatically enforces primary key constraints or generates unique key values, replicating the functionality of a stored procedure.

4. Problem Statement

In enterprise databases, primary keys must be unique for every record. Manual assignment of keys can lead to errors.
Design a trigger that:
Automatically generates or enforces a primary key value during record insertion.

Implements the logic similar to a stored procedure.
Demonstrates automated primary key functionality for a table.
(Industry relevance: Amazon, Flipkart, Oracle)

5. Practical/Experiment Steps

- Identify the table requiring automated primary key enforcement.
- Design a trigger that activates before insert operations.
- Ensure that every new record receives a unique primary key automatically.
- Validate the trigger by inserting multiple records and verifying unique keys.

6. Procedure

- Established a connection to the PostgreSQL server and initialized the working environment using a database client such as pgAdmin.
- Created an **EMPLOYEES** table containing attributes such as employee ID, employee name, gender, and salary.
- Inserted a balanced dataset into the table, ensuring that both male and female employee records were present for meaningful analysis.
- Designed a stored procedure named **GET_EMP_COUNT_BY_GENDER** with properly defined parameters:
  - An **IN parameter** to accept the gender input.
  - An **OUT parameter** to return the total employee count.
  - An **INOUT parameter** to manage and update the execution status.
- Implemented the procedure logic using PL/pgSQL to perform a **COUNT(*) operation**, filtering employee records based on the provided gender value.

- Updated the status variable inside the procedure to **'SUCCESS'** after successful execution of the query.
- Wrote an **anonymous block** to test the procedure, where variables were declared and initialized, including setting the gender input (e.g., 'MALE') and initial status as 'PENDING'.
- Executed the procedure using the **CALL statement**, passing the required parameters and capturing the output values.
- Displayed the results using the **RAISE NOTICE statement**, printing both the employee count and the updated status to the output console.
- Verified the correctness of the output by comparing the returned count with the actual number of records present in the EMPLOYEES table.

7. SQL Queries / Code

```

-- DROP TABLE IF EXISTS (SAFE RE-RUN)

DROP TABLE IF EXISTS EMPLOYEES CASCADE;
DROP SEQUENCE IF EXISTS EMP_SEQ;


-- STEP 1: CREATE TABLE
-- CREATE TABLE
CREATE TABLE EMPLOYEES (

   EMP_ID      INTEGER       PRIMARY KEY,
   EMP_NAME    VARCHAR(100)  NOT NULL,
   DEPARTMENT  VARCHAR(50)   NOT NULL,
   SALARY      NUMERIC(10,2) NOT NULL,
   HIRE_DATE   DATE          DEFAULT CURRENT_DATE

);

-- STEP 2: CREATE SEQUENCE
-- CREATE SEQUENCE FOR PRIMARY KEY GENERATION

CREATE SEQUENCE EMP_SEQ
   START WITH    1
```

```
    INCREMENT BY 1
    NO CYCLE;


-- STEP 3: CREATE TRIGGER FUNCTION
-- TRIGGER FUNCTION TO AUTO-GENERATE PRIMARY KEY

CREATE OR REPLACE FUNCTION FN_EMP_PK()
RETURNS TRIGGER AS $$

BEGIN

    -- CHECK IF EMP_ID IS NULL

    IF NEW.EMP_ID IS NULL THEN
        NEW.EMP_ID := NEXTVAL('EMP_SEQ');
    END IF;

    -- SAFETY CHECK (SHOULD NEVER FAIL)

    IF NEW.EMP_ID IS NULL THEN
        RAISE EXCEPTION 'EMP_ID CANNOT BE NULL — AUTO GENERATION FAILED';
    END IF;

    RETURN NEW;

END;

$$ LANGUAGE PLPGSQL;


-- STEP 4: CREATE TRIGGER
-- ATTACH TRIGGER TO TABLE

CREATE TRIGGER TRG_EMP_PK

BEFORE INSERT ON EMPLOYEES

FOR EACH ROW

EXECUTE FUNCTION FN_EMP_PK();
```

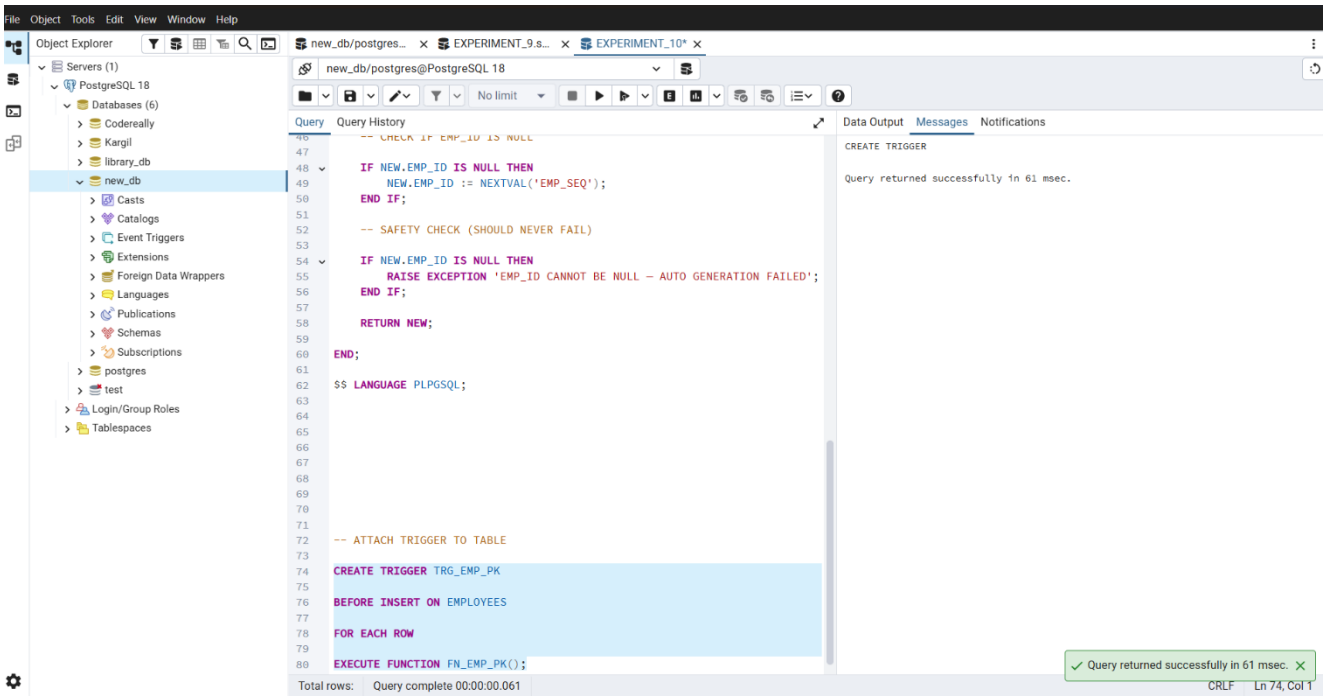
```
-- STEP 5: INSERT TEST RECORDS

-- INSERT WITHOUT PROVIDING EMP_ID

INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('ALICE JOHNSON', 'ENGINEERING', 85000.00);

INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('BOB SMITH', 'MARKETING', 72000.00);

INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('CAROL DAVIS', 'FINANCE', 91000.00);

INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('DAVID LEE', 'ENGINEERING', 78000.00);

INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('EVA MARTINEZ', 'HR', 65000.00);


SELECT * FROM EMPLOYEES


-- STEP 6: VERIFY RESULTS
-- VERIFY AUTO-GENERATED PRIMARY KEYS

SELECT EMP_ID, EMP_NAME, DEPARTMENT, SALARY, HIRE_DATE
FROM EMPLOYEES
ORDER BY EMP_ID;


-- STEP 7: DUPLICATE KEY TEST

-- ATTEMPT TO INSERT DUPLICATE PRIMARY KEY

INSERT INTO EMPLOYEES (EMP_ID, EMP_NAME, DEPARTMENT, SALARY)
VALUES (1, 'DUPLICATE TEST', 'TEST', 50000);
```

```
-- STEP 8: ADMIN / VERIFICATION QUERIES

-- CHECK TRIGGER EXISTS

SELECT TRIGGER_NAME, EVENT_MANIPULATION, ACTION_TIMING
FROM INFORMATION_SCHEMA.TRIGGERS
WHERE EVENT_OBJECT_TABLE = 'EMPLOYEES';

-- CHECK SEQUENCE VALUE

SELECT LAST_VALUE FROM EMP_SEQ;

-- CHECK PRIMARY KEY CONSTRAINT

SELECT CONSTRAINT_NAME, CONSTRAINT_TYPE
FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS
WHERE TABLE_NAME = 'EMPLOYEES';
```

8.Input/Output Analysis

OUTPUT SCREENSHOTS

a:



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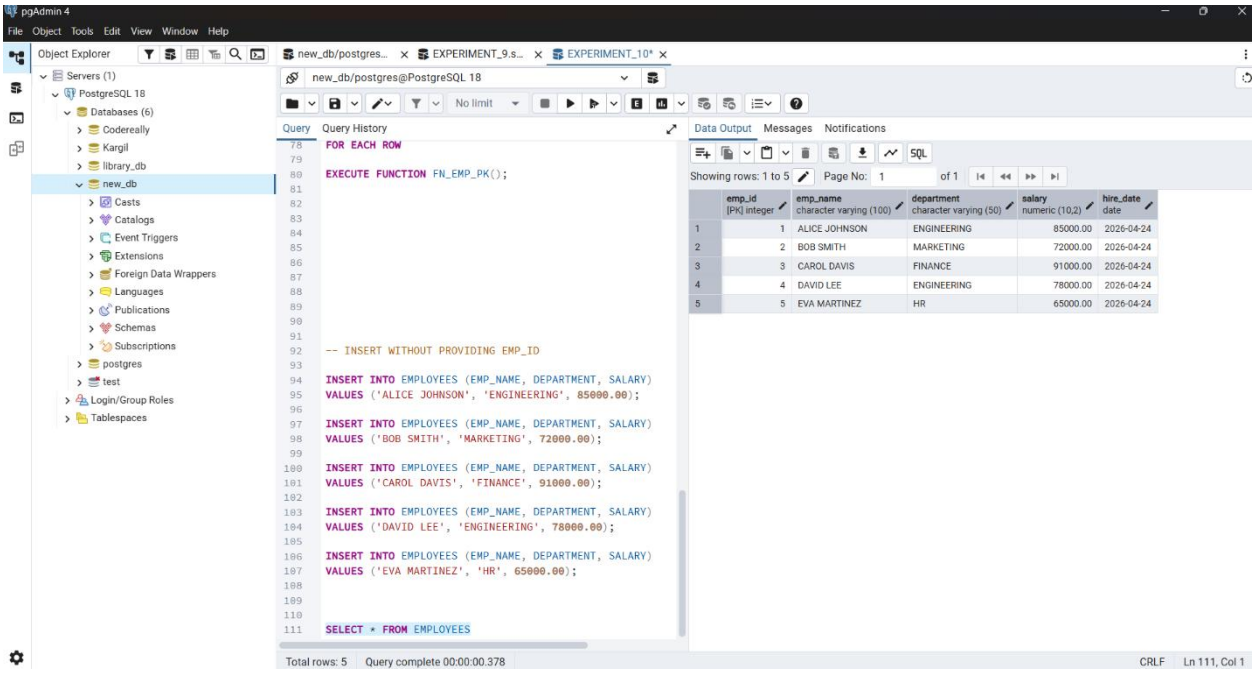
pgAdmin 4
File Object Tools Edit View Window Help
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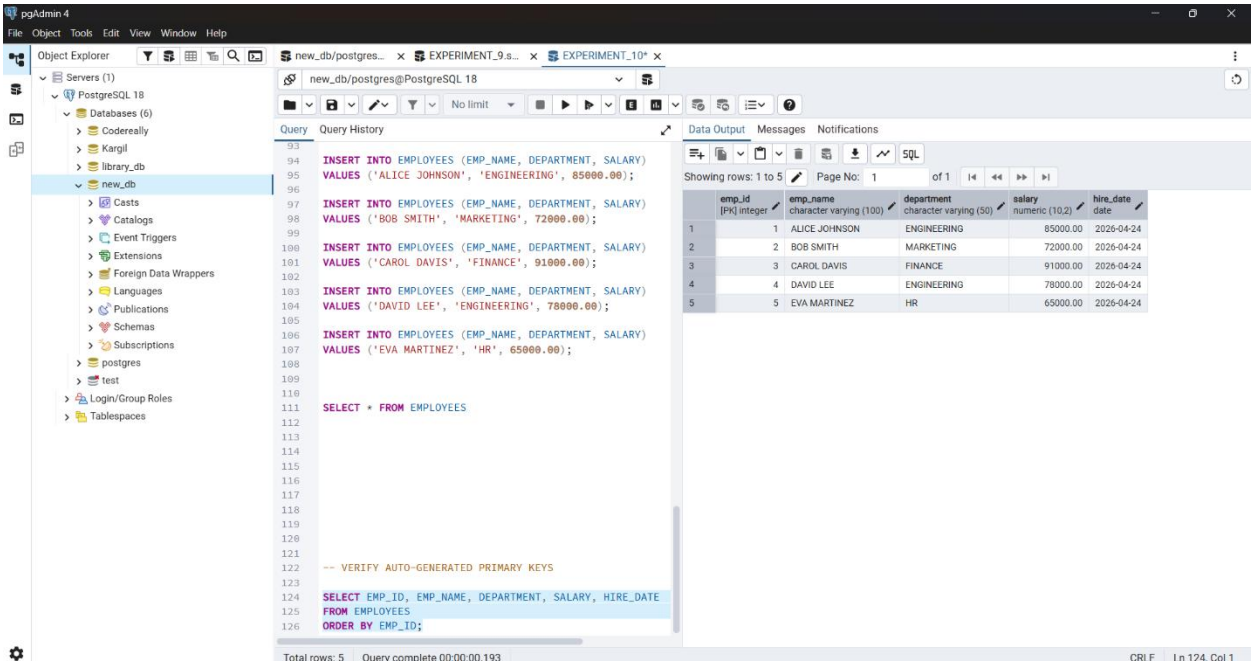
No limit
Query
Query History
FOR EACH ROW
EXECUTE FUNCTION FN_EMP_PK();
-- INSERT WITHOUT PROVIDING EMP_ID
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('ALICE JOHNSON', 'ENGINEERING', 85000.00);
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('BOB SMITH', 'MARKETING', 72000.00);
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('CAROL DAVIS', 'FINANCE', 91000.00);
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('DAVID LEE', 'ENGINEERING', 78000.00);
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('EVA MARTINEZ', 'HR', 65000.00);
SELECT * FROM EMPLOYEES
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 5
Page No: 1
of 1
emp_id [PK] integer
emp_name character varying (100)
department character varying (50)
salary numeric (10,2)
hire_date date
1 1 ALICE JOHNSON ENGINEERING 85000.00 2026-04-24
2 2 BOB SMITH MARKETING 72000.00 2026-04-24
3 3 CAROL DAVIS FINANCE 91000.00 2026-04-24
4 4 DAVID LEE ENGINEERING 78000.00 2026-04-24
5 5 EVA MARTINEZ HR 65000.00 2026-04-24
Total rows: 5
Query complete 00:00:00.378
CRLF
Ln 111, Col 1

c: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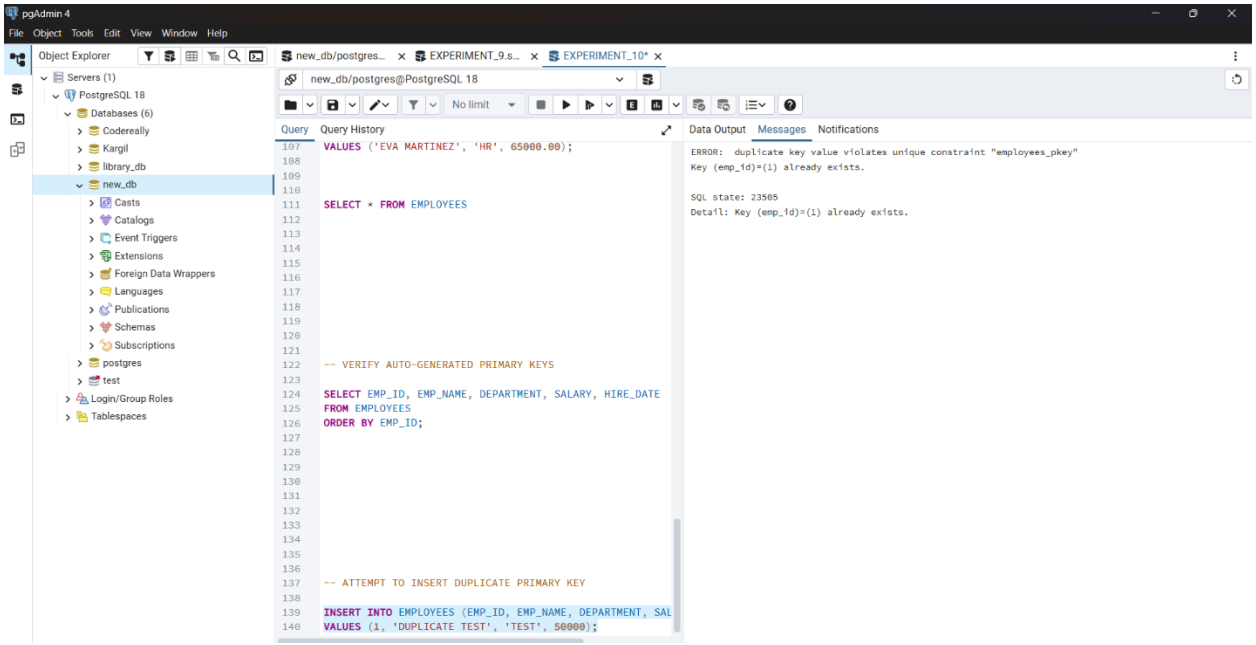
No limit
Query
Query History
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('ALICE JOHNSON', 'ENGINEERING', 85000.00);
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('BOB SMITH', 'MARKETING', 72000.00);
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('CAROL DAVIS', 'FINANCE', 91000.00);
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('DAVID LEE', 'ENGINEERING', 78000.00);
INSERT INTO EMPLOYEES (EMP_NAME, DEPARTMENT, SALARY)
VALUES ('EVA MARTINEZ', 'HR', 65000.00);
SELECT * FROM EMPLOYEES
-- VERIFY AUTO-GENERATED PRIMARY KEYS
SELECT EMP_ID, EMP_NAME, DEPARTMENT, SALARY, HIRE_DATE
FROM EMPLOYEES
ORDER BY EMP_ID;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 5
Page No: 1
of 1
emp_id [PK] integer
emp_name character varying (100)
department character varying (50)
salary numeric (10,2)
hire_date date
1 1 ALICE JOHNSON ENGINEERING 85000.00 2026-04-24
2 2 BOB SMITH MARKETING 72000.00 2026-04-24
3 3 CAROL DAVIS FINANCE 91000.00 2026-04-24
4 4 DAVID LEE ENGINEERING 78000.00 2026-04-24
5 5 EVA MARTINEZ HR 65000.00 2026-04-24
Total rows: 5
Query complete 00:00:00.193
CRLF
Ln 124, Col 1

d: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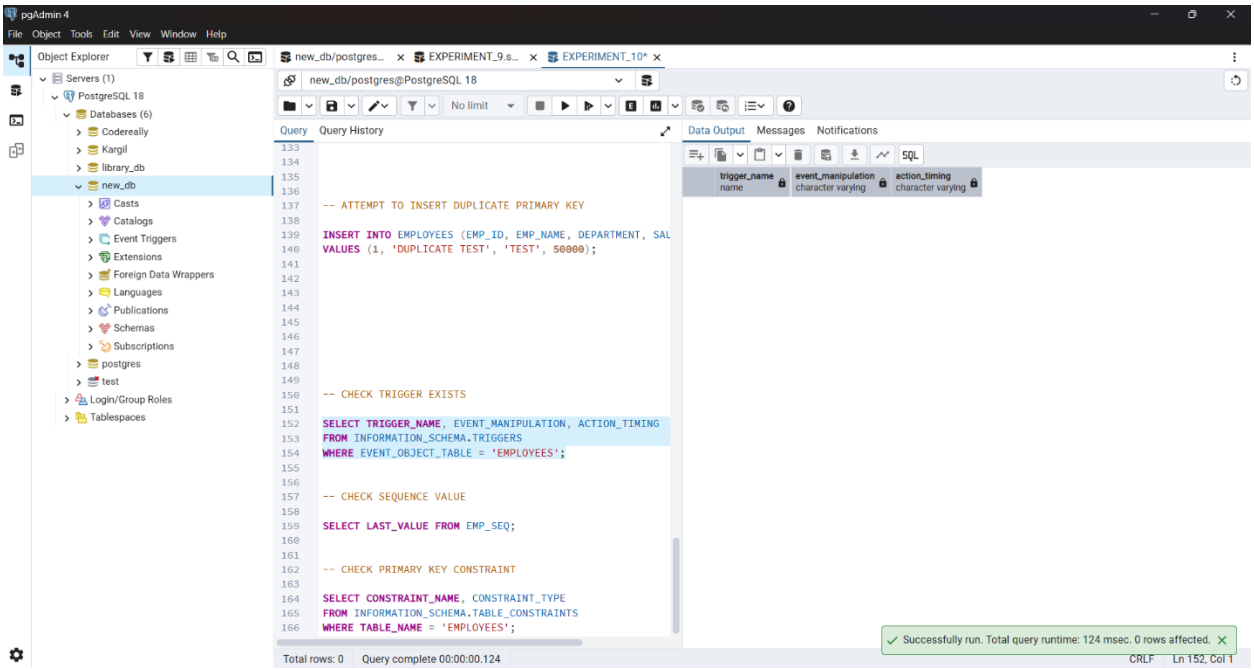
No limit
Query
Query History
VALUES ('EVA MARTINEZ', 'HR', 65000.00);
SELECT * FROM EMPLOYEES
-- VERIFY AUTO-GENERATED PRIMARY KEYS
SELECT EMP_ID, EMP_NAME, DEPARTMENT, SALARY, HIRE_DATE
FROM EMPLOYEES
ORDER BY EMP_ID;
-- ATTEMPT TO INSERT DUPLICATE PRIMARY KEY
INSERT INTO EMPLOYEES (EMP_ID, EMP_NAME, DEPARTMENT, SAL
VALUES (1, 'DUPLICATE TEST', 'TEST', 50000);
Data Output
Messages
Notifications
ERROR: duplicate key value violates unique constraint "employees_pkey"
Key (emp_id)=(1) already exists.
SQL state: 23505
Detail: Key (emp_id)=(1) already exists.

e: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EXPERIMENT_10*
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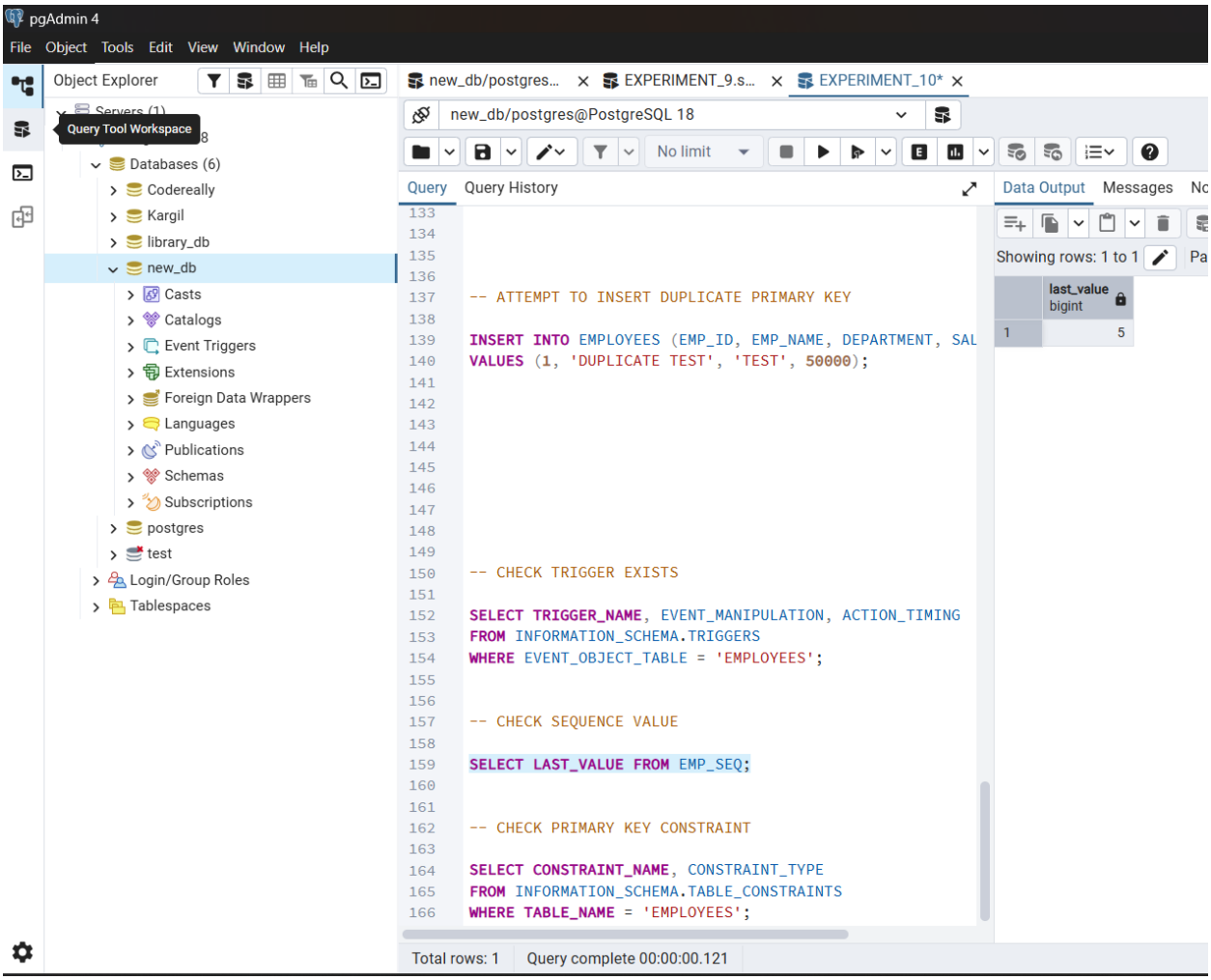
No limit
Query
Query History
-- ATTEMPT TO INSERT DUPLICATE PRIMARY KEY
INSERT INTO EMPLOYEES (EMP_ID, EMP_NAME, DEPARTMENT, SAL
VALUES (1, 'DUPLICATE TEST', 'TEST', 50000);
-- CHECK TRIGGER EXISTS
SELECT TRIGGER_NAME, EVENT_MANIPULATION, ACTION_TIMING
FROM INFORMATION_SCHEMA.TRIGGERS
WHERE EVENT_OBJECT_TABLE = 'EMPLOYEES';
-- CHECK SEQUENCE VALUE
SELECT LAST_VALUE FROM EMP_SEQ;
-- CHECK PRIMARY KEY CONSTRAINT
SELECT CONSTRAINT_NAME, CONSTRAINT_TYPE
FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS
WHERE TABLE_NAME = 'EMPLOYEES';
Data Output
Messages
Notifications
SQL
trigger_name name
event_manipulation character varying
action_timing character varying
Successfully run. Total query runtime: 124 msec. 0 rows affected.
Total rows: 0
Query complete 00:00:00.124
CRLF
Ln 152, Col 1

f: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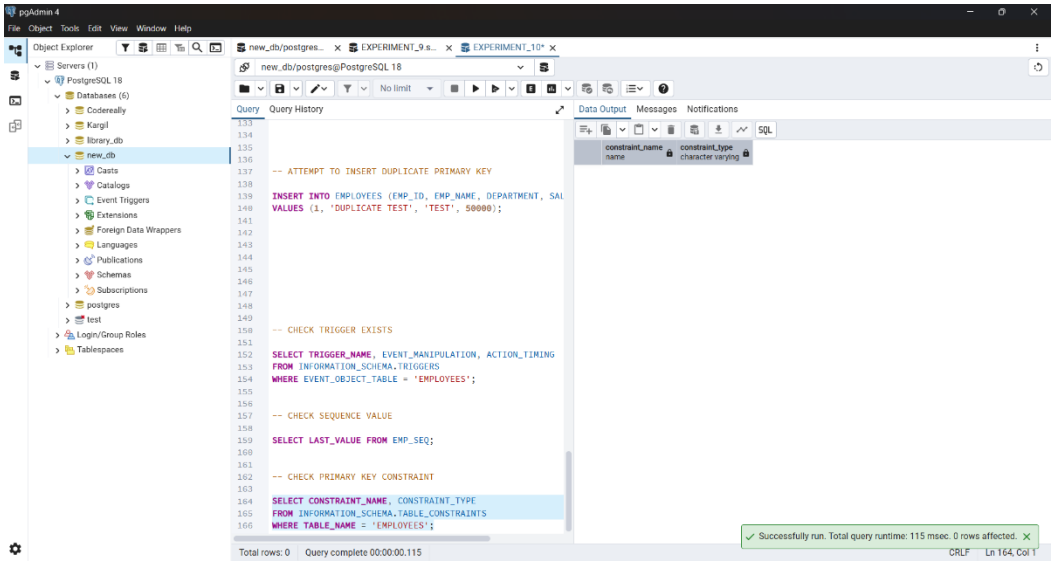
EXPERIMENT_9.s...
EXPERIMENT_10*
new_db/postgres@PostgreSQL 18
No limit
Query
Query History
-- ATTEMPT TO INSERT DUPLICATE PRIMARY KEY
INSERT INTO EMPLOYEES (EMP_ID, EMP_NAME, DEPARTMENT, SAL
VALUES (1, 'DUPLICATE TEST', 'TEST', 50000);
-- CHECK TRIGGER EXISTS
SELECT TRIGGER_NAME, EVENT_MANIPULATION, ACTION_TIMING
FROM INFORMATION_SCHEMA.TRIGGERS
WHERE EVENT_OBJECT_TABLE = 'EMPLOYEES';
-- CHECK SEQUENCE VALUE
SELECT LAST_VALUE FROM EMP_SEQ;
-- CHECK PRIMARY KEY CONSTRAINT
SELECT CONSTRAINT_NAME, CONSTRAINT_TYPE
FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS
WHERE TABLE_NAME = 'EMPLOYEES';
Data Output
Messages
Showing rows: 1 to 1
last_value
bigint
1
5
Total rows: 1
Query complete 00:00:00.121

g: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No limit
Query
Query History
-- ATTEMPT TO INSERT DUPLICATE PRIMARY KEY
INSERT INTO EMPLOYEES (EMP_ID, EMP_NAME, DEPARTMENT, SAL
VALUES (1, 'DUPLICATE TEST', 'TEST', 50000);
-- CHECK TRIGGER EXISTS
SELECT TRIGGER_NAME, EVENT_MANIPULATION, ACTION_TIMING
FROM INFORMATION_SCHEMA.TRIGGERS
WHERE EVENT_OBJECT_TABLE = 'EMPLOYEES';
-- CHECK SEQUENCE VALUE
SELECT LAST_VALUE FROM EMP_SEQ;
-- CHECK PRIMARY KEY CONSTRAINT
SELECT CONSTRAINT_NAME, CONSTRAINT_TYPE
FROM INFORMATION_SCHEMA.TABLE_CONSTRAINTS
WHERE TABLE_NAME = 'EMPLOYEES';
Data Output
Messages
Notifications
constraint_name
name
constraint_type
character varying
Successfully run. Total query runtime: 115 msec. 0 rows affected.
Total rows: 0
Query complete 00:00:00.115
CRLF
Ln 164, Col 1

9. Learning Outcomes

- After completing this experiment, the learner will be able to:
- Understand the purpose and working of database triggers.
- Implement automated primary key functionality using triggers.
- Ensure data integrity without manual key assignment.
- Apply trigger-based automation in real-world enterprise applications such as Amazon, Flipkart, and Oracle.